

Table of Contents

Executive Summary.....	II
Systems and Procedures.....	1
Results.....	2
Conclusion.....	3
Appendix 1.....	4
Appendix 2.....	6
Appendix 3.....	10
Appendix 4.....	15
Appendix 5.....	17
References.....	20

Executive Summary

In this problem we deal with the liquefaction of natural gas via the Claude process. The liquification of natural gas involves a three-stage compression system, two heat exchangers, an expander, and a flash tank. After setting up our PFD for this process using the four major equipment listed above, a heat and weight balance was obtained using Aspen stream summary. Furthermore, TEMA sheets are obtained for the two heat exchangers, the precooler and the two inter-stage coolers using ASPEN Plus EDR. Knowing this, we can calculate our total annual profit after the cost of the electricity and equipment. This annual profit for the Claude process was compared to the Linde process. Lastly, using sensitivity analysis in Aspen we found the optimal flow into the expander.

Systems and Procedures

To begin our process a feed stream of methane, ethane, propane, n-butane, and nitrogen operating at 20° C and 1 bar with a total flow rate of 25 kmol/hr was the inlet stream going into a multi-stage compressor. Looking at our given assumptions, in our multi-stage compressor we inputted that it is a 3-stage compressor operating at 85% efficiency and the discharge pressure of the third stage is 30 bar. Next, we made our stream out the compressor go into a pre-cooler where there is an allowable pressure drop of 10 psi and cooling water is available at 30° C. The stream out the cooler entered our first heat exchanger where there is an allowable pressure drop of 10 psi as well. Next, our stream out was then split where one stream went into the second heat exchanger and another stream went to the expander. We had to add a constraint for the outlet temperature of the cold stream entering the second heat exchanger to be at 3° C. The same approach was used regarding the pressure drop for the second heat exchanger and expander was inputted to have an 85% efficiency. A design specification was added to ensure that the flowrate of the expander is 35% of the flow of the hot stream entering the first heat exchanger. Our stream out of the second heat exchanger went into a pressure valve and then into a flash tank. Evidently the top stream out the splitter would go into a mixer with an outlet stream of the expander. The outlet stream of this mixer which would be the recycle stream and would be recycled back into the fresh feed stream that enters the multi-stage compressor. The process was able to converge to LNG production rate of 413 kg/hr. The total equipment cost of the four major equipment were calculated using the equation in Appendix 1. Given our operating cost of electricity we can calculate our total operating cost. Using these values, we obtained our total annualized profit for LNG production using the Claude process.

Results

1. Problem 6A – Liquification of Natural Gas

- i. With the given data, we were able to design another aspen simulation for the liquification of natural gas. The heat and weight balances, TEMA sheets for the given major equipment , and annual profit.
- ii. Additional Data:
 - i. Equations and Formulas – See *Appendix 1*
 - ii. PFD – See *Appendix 2*
 - iii. Material Stream Data – See *Appendix 3*
 - iv. Heat Data – See *Appendix 3*
 - v. TEMA sheets- See *Appendix 4*
 - vi. Capital, Operating, and Annual Information – See *Appendix 5*

Conclusion

In conclusion for this problem we were able to design a PFD for the liquification of natural gas via the Claude process. We obtained all necessary stream summaries and heat and weight balances. Furthermore, using Aspen we obtained the TEMA sheets for the necessary equipment. Doing this we see the actual cost which includes the installation/manpower cost. Knowing our major equipment and our utilities, we obtained the capital and operating cost. Knowing this, we found the annual profit for the liquification of natural gas and compared this to the Linde process to see which was, the more efficient process. Lastly, we found the optimal flow to the expander using sensitivity analysis that gave us the most efficient process with the highest annual profit.

Appendix 1

Equations and Formulas

Shell and Tube Heat Exchangers:

$$\text{Purchase Cost (2010)} = 28,000 + 54 * A^{1.2}$$

$$\text{Purchase Cost (2020)} = \text{Purchase Cost (2010)} + \frac{\text{index}(2020)}{\text{index}(2010)}$$

$$\text{Where: } \text{index}(2010) = 1500$$

$$\text{To obtain 'A': } Q = F_t * A * U * \Delta T_{ln}$$

$$\text{Where: } Q = \text{Obtained from Aspen Simulation}$$

$$F_t = 0.8$$

$$U = 50 \frac{\text{Btu}}{\text{hr} * \text{ft}^2 * ^\circ\text{F}}$$

$$\Delta T_{ln} = \frac{\Delta T_1 - \Delta T_2}{\ln (\Delta T_1 / \Delta T_2)}$$

$$\text{Installation Factor} = 3.5$$

$$\text{Installation Cost} = 3.5 * \text{Purchase Cost (2020)}$$

$$\text{Annualized Cost} = \frac{4}{3} * \text{Installation Cost}$$

Compressors:

$$\text{Purchase Cost (2010)} = 500,000 + 20,000 * W^{0.6}$$

$$\text{Purchase Cost (2020)} = \text{Purchase Cost (2010)} + \frac{\text{index}(2020)}{\text{index}(2010)}$$

$$\text{Where: } \text{index}(2010) = 1500$$

$$W = \text{Obtained from Aspen Simulation in Kw.}$$

$$\text{Installation Factor} = 2.5$$

$$\text{Installation Cost} = 2.5 * \text{Purchase Cost (2020)}$$

$$\text{Annualized Cost} = \frac{4}{3} * \text{Installation Cost}$$

Drums/Flash Tank:

Assumptions: $\frac{L}{D} = 2$, Residence Time= 5 minutes

$$\text{Purchase Cost (2010)} = 11,600 + 34 * M^{0.85}$$

$$\text{Purchase Cost (2020)} = \text{Purchase Cost (2010)} + \frac{\text{index}(2020)}{\text{index}(2010)}$$

Where: $\text{index}(2010) = 1500$

Where: $M = (\pi * D * L) * \text{Thickness} * \rho_{\text{steel}}$

$$\rho_{\text{steel}} = 8000 \frac{\text{kg}}{\text{m}^3}$$

$$\text{Thickness} = \frac{P_i D_i}{2 * S * E - 1.2 * P_i}$$

Where: E= Welded Efficiency= 1

$$S_{\text{steel}} = \text{Stress} = 12.9 \text{ Ksi}$$

P_i = Internal Operating Pressure (from Aspen Simulation plus 10% safety factor)

D_i = Internal Diameter

$$\text{Where: } D_i = \frac{2 * (V_{\text{vessel}})^{1/3}}{\pi}$$

Installation Factor= 4

Installation Cost= 4 * Purchase Cost (2020)

Annualized Cost= $\frac{4}{3}$ * Installation Cost

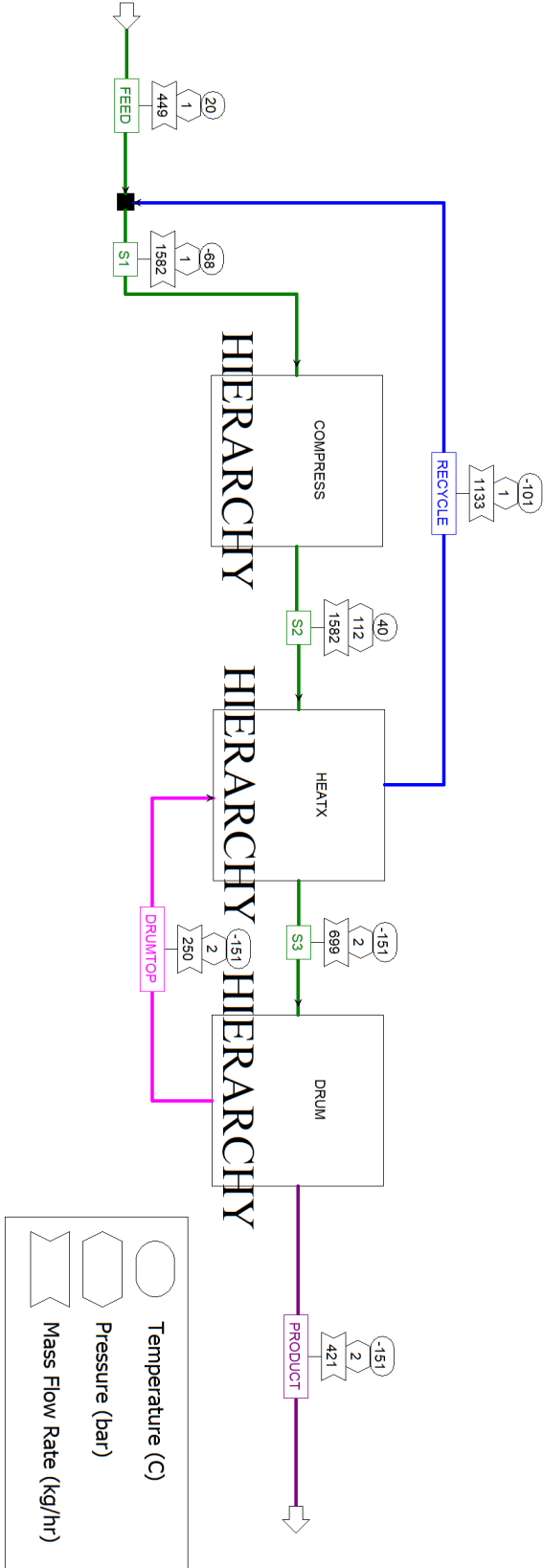


Figure 1 – Overall PFD

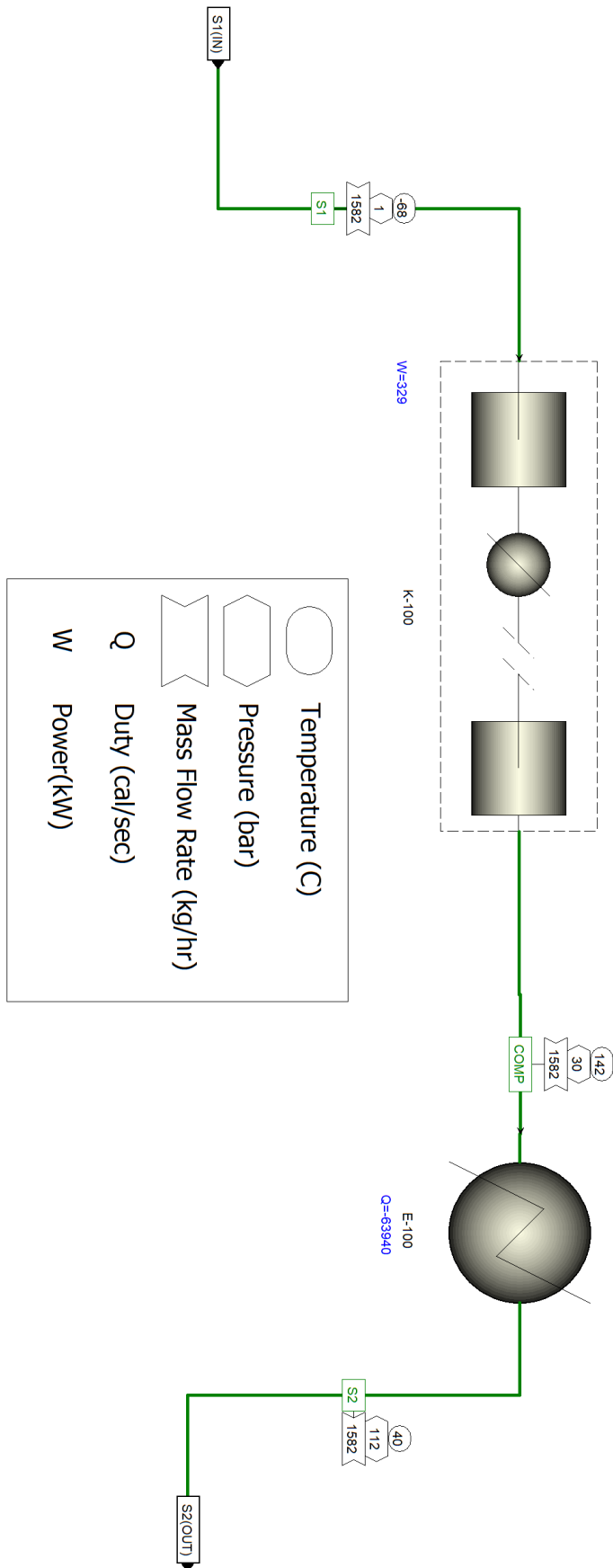


Figure 2 – Compress Hierarchy PFD

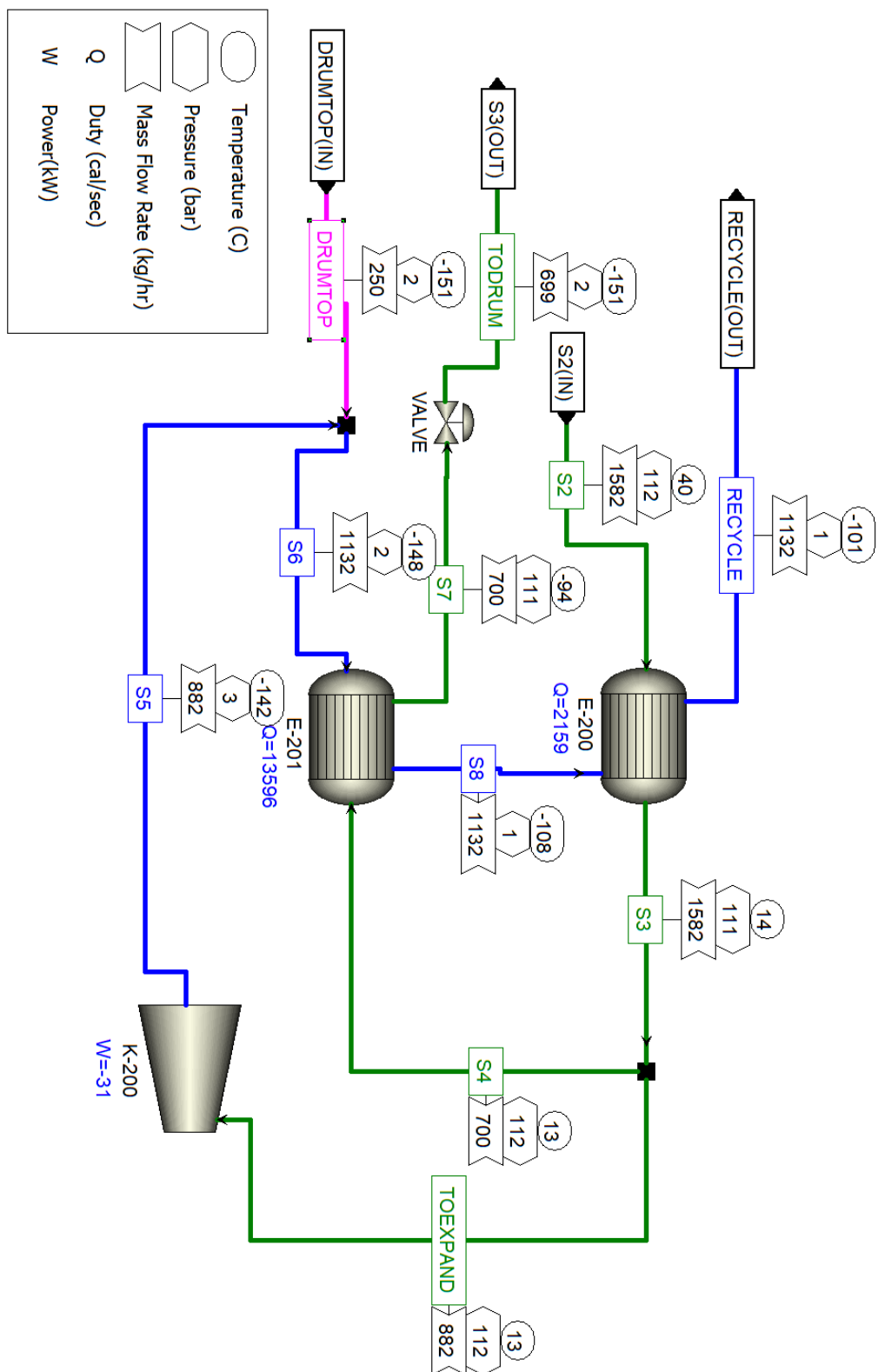


Figure 3 – Heatx Hierarchy PFD

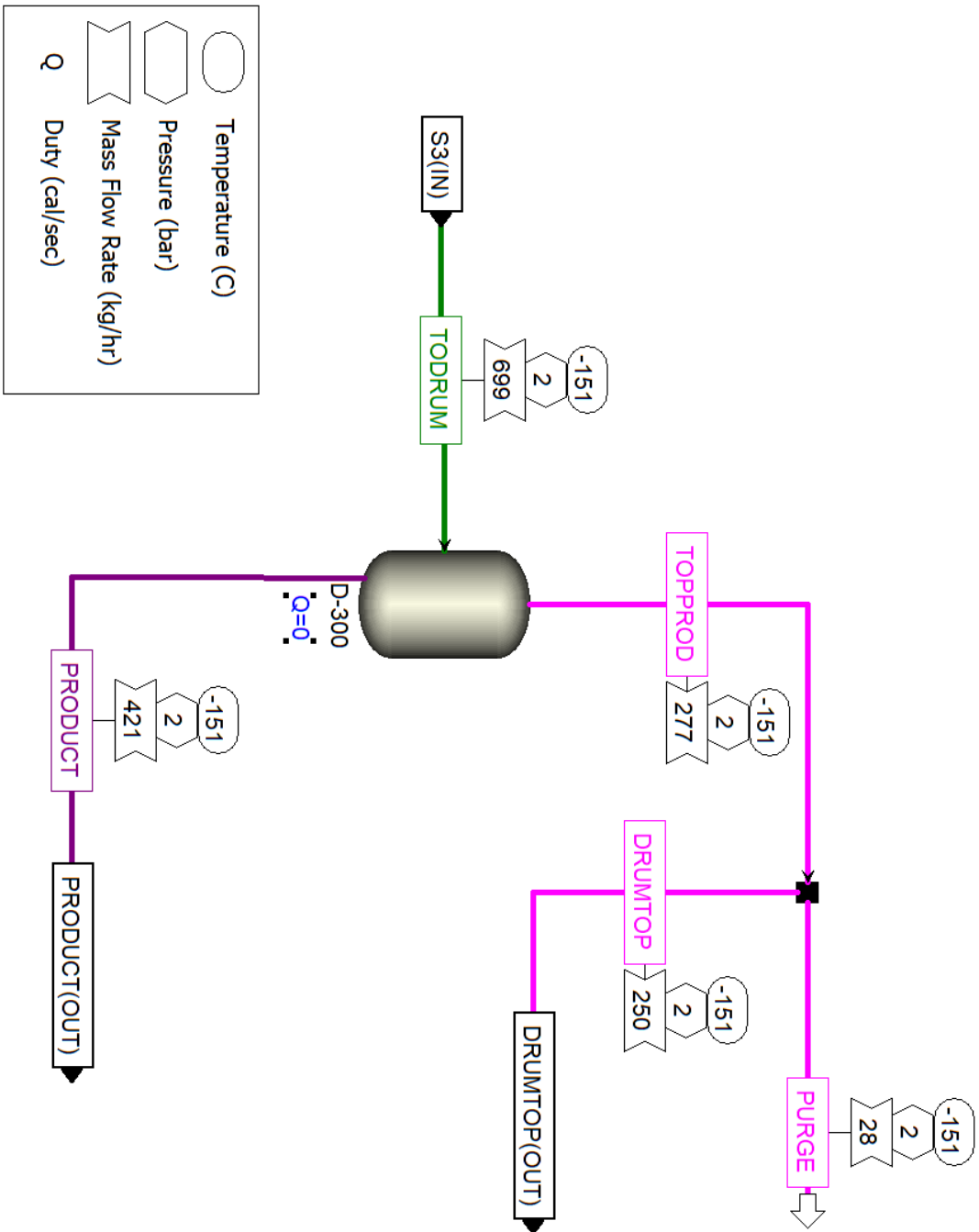


Figure 4 – Drum Hierarchy PFD

Table 1 – Material Stream Data

Stream Name	Units	COMPRESS.COMP	COMPRESS.S1	COMPRESS.S2	DRUM.DRUMTOP	DRUM.PRODUCT	DRUM.PURGE	DRUM.TODRUM	DRUM.TOPPROD
Description									
From		COMPRESS.K-100	\$C-1	COMPRESS.E-100					
To		COMPRESS.E-100	COMPRESS.K-100	\$C-2	DRUM.B2	DRUM.D-300	DRUM.B2	\$C-6	DRUM.D-300
Phase		Vapor	Vapor	Liquid	Vapor	Liquid	Vapor	Mixed	Vapor
Temperature	C	141.61	-67.767	40	-150.917	-150.917	-150.917	-150.917	-150.917
Pressure	bar	30	0.621049	112.554	2	2	2	2	2
Mole Fractions									
METHANE		0.937914	0.937914	0.937914	0.99437	0.896059	0.99437	0.937911	0.99437
ETHANE		0.0391096	0.0391096	0.0391096	0.00018882	0.0679722	0.00018882	0.039116	0.00018882
PROPANE		0.0153563	0.0153563	0.0153563	7.56E-07	0.0267439	7.56E-07	0.015359	7.56E-07
N-BUTANE		0.00493366	0.00493366	0.00493366	2.75E-09	0.00859245	2.75E-09	0.00493454	2.75E-09
NITROGEN		0.00268652	0.00268652	0.00268652	0.00544064	0.000632716	0.00544064	0.00267951	0.00544064
Mass Flows	kg/hr	1582.18	1582.18	1582.18	250.206	421.707	27.8007	699.714	278.007
METHANE	kg/hr	1379.14	1379.14	1379.14	247.75	334.636	27.5278	609.915	275.278
ETHANE	kg/hr	107.79	107.79	107.79	0.0881788	47.5792	0.00979764	47.6772	0.0979764
PROPANE	kg/hr	62.0664	62.0664	62.0664	0.000517505	27.4528	5.75E-05	27.4534	0.000575006
N-BUTANE	kg/hr	26.2837	26.2837	26.2837	2.48E-06	11.6259	2.75E-07	11.6259	2.75E-06
NITROGEN	kg/hr	6.89801	6.89801	6.89801	2.36703	0.412604	0.263004	3.04264	2.63004

Table 2 – Material Stream Data (Cont.)

Stream Name	Units	DRUMTOP	FEED	HEATX.DRUMTOP	HEATX.RECYCLE	HEATX.S2	HEATX.S3	HEATX.S4	HEATX.S5
Description									
From		\$C-7		\$C-8	HEATX.E-200	\$C-3	HEATX.E-200	HEATX.B2	HEATX.K-200
To		\$C-8	B1	HEATX.B4	\$C-4	HEATX.E-200	HEATX.B2	HEATX.E-201	HEATX.B4
Phase		Vapor	Vapor	Vapor	Mixed	Liquid	Liquid	Liquid	Mixed
Temperature	C	-150.917	20	-150.917	-100.518	40	13.6	13.4203	-141.795
Pressure	bar	2	1	2	0.621049	112.554	111.864	112	3
Mole Fractions									
METHANE		0.99437	0.9028	0.99437	0.951063	0.937914	0.937914	0.937904	0.937904
ETHANE		0.00018882	0.0633	0.00018882	0.0300445	0.0391096	0.0391096	0.0391165	0.0391165
PROPANE		7.56E-07	0.0249	7.56E-07	0.0117797	0.0153563	0.0153563	0.0153589	0.0153589
N-BUTANE		2.75E-09	0.008	2.75E-09	0.00378452	0.00493366	0.00493366	0.00493449	0.00493449
NITROGEN		0.00544064	0.001	0.00544064	0.00332836	0.00268652	0.00268652	0.00268651	0.00268651
Mass Flows	kg/hr	250.206	449.445	250.206	1132.48	1582.18	1582.18	699.678	882.27
METHANE	kg/hr	247.75	362.085	247.75	1016.78	1379.14	1379.14	609.875	769.032
ETHANE	kg/hr	0.0881788	47.5852	0.0881788	60.205	107.79	107.79	47.6752	60.1168
PROPANE	kg/hr	0.000517505	27.4501	0.000517505	34.6161	62.0664	62.0664	27.4516	34.6156
N-BUTANE	kg/hr	2.48E-06	11.6247	2.48E-06	14.6589	26.2837	26.2837	11.6251	14.6589
NITROGEN	kg/hr	2.36703	0.700337	2.36703	6.21351	6.89801	6.89801	3.05042	3.84647

Table 3 – Material Stream Data (Cont.)

Stream Name	Units	HEATX.S6	HEATX.S7	HEATX.S8	HEATX.TODRUM	HEATX.TOEXPAND	PRODUCT	RECYCLE	S1
Description									
From		HEATX.B4	HEATX.E-201	HEATX.E-201	HEATX.VALVE	HEATX.B2	\$C-9	\$C-4	B1
To		HEATX.E-201	HEATX.VALVE	HEATX.E-200	\$C-5	HEATX.K-200		B1	\$C-1
Phase		Mixed	Liquid	Mixed	Mixed	Liquid	Liquid	Mixed	Vapor
Temperature	C	-147.953	-94	-108.216	-150.917	13.4203	-150.917	-100.535	-67.767
Pressure	bar	2	111.311	1.31052	2	112	2	0.621049	0.621049
Mole Fractions									
METHANE		0.951063	0.937904	0.951063	0.937911	0.937904	0.896059	0.951084	0.937914
ETHANE		0.0300445	0.0391165	0.0300445	0.039116	0.0391165	0.0679722	0.0300369	0.0391096
PROPANE		0.0117797	0.0153589	0.0117797	0.015359	0.0153589	0.0267439	0.0117769	0.0153563
N-BUTANE		0.00378452	0.00493449	0.00378452	0.00493454	0.00493449	0.00859245	0.00378361	0.00493366
NITROGEN		0.00332836	0.00268651	0.00332836	0.00267951	0.00268651	0.000632716	0.00331906	0.00268652
Mass Flows	kg/hr	1132.48	699.678	1132.48	699.714	882.27	421.707	1132.73	1582.18
METHANE	kg/hr	1016.78	609.875	1016.78	609.915	769.032	334.636	1017.06	1379.14
ETHANE	kg/hr	60.205	47.6752	60.205	47.6772	60.1168	47.5792	60.2046	107.79
PROPANE	kg/hr	34.6161	27.4516	34.6161	27.4534	34.6156	27.4528	34.6164	62.0664
N-BUTANE	kg/hr	14.6589	11.6251	14.6589	11.6259	14.6589	11.6259	14.659	26.2837
NITROGEN	kg/hr	6.21351	3.05042	6.21351	3.04264	3.84647	0.412604	6.19767	6.89801

Table 4 – Material Stream Data (Cont.)

Stream Name	Units	S2	S3
Description			
From		\$C-2	\$C-5
To		\$C-3	\$C-6
Phase		Liquid	Mixed
Temperature	C	40	-150.917
Pressure	bar	112.554	2
Mole Fractions			
METHANE		0.937914	0.937911
ETHANE		0.0391096	0.039116
PROPANE		0.0153563	0.015359
N-BUTANE		0.00493366	0.00493454
NITROGEN		0.00268652	0.00267951
Mass Flows	kg/hr	1582.18	699.714
METHANE	kg/hr	1379.14	609.915
ETHANE	kg/hr	107.79	47.6772
PROPANE	kg/hr	62.0664	27.4534
N-BUTANE	kg/hr	26.2837	11.6259
NITROGEN	kg/hr	6.89801	3.04264

Table 5 – Heat Data

E-100	E-200	E-201
cal/sec	cal/sec	cal/sec
-63877.61	2159.84	13598.07

Table 6 – Detailed 3-stage Compressor Data

Stage	Temperature C	Pressure bar	Pressure ratio	Indicated power kW	Brake horsepower kW	Head developed m-kgf/kg	Volumetric flow l/min	Efficiency used
1	17.8251462	2.26181692	3.64193259	77.000684	77.000684	15185.8761	41824.5301	0.85
2	125.551295	8.23738476	3.64193259	107.560847	107.560847	21212.8725	16253.4093	0.85
3	253.857466	30	3.64193259	145.996463	145.996463	28793.0454	6182.49313	0.85

Appendix 4

Aspen Exchanger Design and Rating Shell & Tube V9

File:

Printed: 12/2/2019 at 6:02:04 PM

TEMA Sheet

Heat Exchanger Specification Sheet										
1	Company:									
2	Location:									
3	Service of Unit:				Our Reference:					
4	Item No.:				Your Reference:					
5	Date:		Rev No.:		Job No.:					
6	Size: 8.0709 - 47.2441		in		Type: BEM Horizontal		Connected in: 1 parallel 1 series			
7	Surf/unit(eff.)		15.7		ft ²		Shells/unit 1		Surf/shell(eff.) 15.7 ft ²	
8	PERFORMANCE OF ONE UNIT									
9	Fluid allocation				Shell Side			Tube Side		
10	Fluid name				HEATX.S8			HEATX.S2		
11	Fluid quantity, Total				lb/h			2020 3011		
12	Vapor (In/Out)				lb/h			1916 1983 0 0		
13	Liquid				lb/h			104 37 3011 3011		
14	Noncondensable				lb/h			0 0 0 0		
15										
16	Temperature (In/Out)				°F			-145.97 -123.91 104 55.96		
17	Dew / Bubble point				°F			-109.97 -251.11 118.79 103.98		
18	Density Vapor/Liquid				lb/ft ³			0.099 / 40.242 0.088 / 40.974 / 10.497 / 10.497		
19	Viscosity				cp			0.0068 / 0.3932 0.0072 / 0.4117 / 0.0019 / 0.0029		
20	Molecular wt, Vap							16.39 17.39		
21	Molecular wt, NC									
22	Specific heat				BTU/(lb-F)			0.4843 / 0.5586 0.479 / 0.6638 / 0.4096 / 0.1544		
23	Thermal conductivity				BTU/(ft-h-F)			0.01 / 0.093 0.011 / 0.089 / 0.045 / 0.046		
24	Latent heat				BTU/lb			207.3 194.9		
25	Pressure (abs)				psi			19.01 17.94 1557.68 1556.43		
26	Velocity (Mean/Max)				ft/s			35.43 / 54.5 7.85 / 8.57		
27	Pressure drop, allow./calc.				psi			10 1.07 10 1.25		
28	Fouling resistance (min)				ft ² -h-F/BTU			0 0 0 Ao based		
29	Heat exchanged				BTU/h			35121 MTD (corrected) 226.59 °F		
30	Transfer rate, Service				9.9 Dirty 63.21 Clean 63.21 BTU/(h-ft ² -F)					
31	CONSTRUCTION OF ONE SHELL									
32					Shell Side			Tube Side		
33	Design/Vacuum/test pressure				psi			43.51 / / 1725.95 / /		
34	Design temperature				°F			167 167		
35	Number passes per shell							1 4		
36	Corrosion allowance				in			0.125 0.125		
37	Connections				In in 1 4 / - 1 1 / -					
38	Size/Rating				Out 1 3 / - 1 1.25 / -					
39	Nominal				Intermediate / - / -					
40	Tube No. 22 OD 0.75 TksAverage 0.083				in Length 47.2441			in Pitch 0.9375 in		
41	Tube type Plain #/in Material Carbon Steel							Tube pattern 30		
42	Shell Carbon Steel ID 8.071 OD 8.625				in			Shell cover -		
43	Channel or bonnet Carbon Steel							Channel cover -		
44	Tubesheet-stationary Carbon Steel -							Tubesheet-floating -		
45	Floating head cover -							Impingement protection None		
46	Baffle-cross Carbon Steel Type Single segmental Cut(%d) 41.48				VertiSpacing: c/c 5.315			in		
47	Baffle-long - Seal Type							Inlet 8.4636 in		
48	Supports-tube U-bend 0				Type					
49	Bypass seal				Tube-tubesheet joint Expanded only (2 grooves)(App.A 'i')					
50	Expansion joint -				Type None					
51	RhoV2-Inlet nozzle 388				Bundle entrance 148			Bundle exit 211 lb/(ft-s ²)		
52	Gaskets - Shell side -				Tube side			Flat Metal Jacket Fibe		
53	Floating head -									
54	Code requirements ASME Code Sec VIII Div 1				TEMA class R - refinery service					
55	Weight/Shell 610.1 Filled with water 693.3				Bundle 158.8			lb		
56	Remarks									
57										
58										

Figure 5 – TEMA Sheet for E-200

Aspen Exchanger Design and Rating Shell & Tube V9

File:

Printed: 12/2/2019 at 6:04:40 PM

TEMA Sheet

Heat Exchanger Specification Sheet									
1	Company:								
2	Location:								
3	Service of Unit:				Our Reference:				
4	Item No.:				Your Reference:				
5	Date:	Rev No.:	Job No.:						
6	Size: 2.0866 - 212.598	in	Type:	BEM	Horizontal	Connected in:	1	parallel	1 series
7	Surf/unit(eff.)	339.4	ft ²	Shells/unit	1	Surf/shell(eff.)	339.4	ft ²	
8	PERFORMANCE OF ONE UNIT								
9	Fluid allocation			Shell Side			Tube Side		
10	Fluid name			HEATX.S6			HEATX.S4		
11	Fluid quantity, Total			2020			1066		
12	Vapor (In/Out)			1380			1906		
13	Liquid			641			114		
14	Noncondensable			0			0		
15									
16	Temperature (In/Out)			°F -230.39			°F -143.15		
17	Dew / Bubble point			°F -99.87			°F -240.46		
18	Density Vapor/Liquid			lb/ft ³ 0.199 / 32.085			lb/ft ³ 0.125 / 39.783		
19	Viscosity			cp 0.0051 / 0.1688			cp 0.0069 / 0.3563		
20	Molecular wt, Vap			16.22			16.87		
21	Molecular wt, NC								
22	Specific heat			BTU/(lb-F) 0.5212 / 0.716			BTU/(lb-F) 0.4871 / 0.5749		
23	Thermal conductivity			BTU/(ft-h-F) 0.008 / 0.105			BTU/(ft-h-F) 0.01 / 0.091		
24	Latent heat			BTU/lb 207.2			BTU/lb 202.1		
25	Pressure (abs)			psi 29.01			psi 24.28		
26	Velocity (Mean/Max)			ft/s 39.63 / 80.9			ft/s 0.11 / 0.15		
27	Pressure drop, allow./calc.			psi 10			psi 4.72		
28	Fouling resistance (min)			ft ² -h-F/BTU 0			ft ² -h-F/BTU 0		
29	Heat exchanged			202822 BTU/h			MTD (corrected) 45.36 °F		
30	Transfer rate, Service			13.17 Dirty			13.23 Clean		
31	CONSTRUCTION OF ONE SHELL								
32				Shell Side			Tube Side		
33	Design/Vacuum/test pressure			psi 43.51 /			psi 1798.47 /		
34	Design temperature			°F 113			°F 113		
35	Number passes per shell			1			1		
36	Corrosion allowance			in 0.125			in 0.125		
37	Connections			In 1 3 / -			In 1 0.5 / -		
38	Size/Rating			Out 1 3 / -			Out 1 0.5 / -		
39	Nominal			Intermediate / -			Intermediate / -		
40	Tube No. 100			OD 0.75			TksAverage 0.083		
41	Tube type Plain			#/in			Material Carbon Steel		
42	Shell Carbon Steel			ID 12.09			in Shell cover		
43	Channel or bonnet Carbon Steel						Channel cover		
44	Tubesheet-stationary Carbon Steel						Tubesheet-floating		
45	Floating head cover						Impingement protection		
46	Baffle-cross Carbon Steel			Type Single segmental			Cut(%d) 23.14		
47	Baffle-long			Seal Type			Inlet 6.8691		
48	Supports-tube U-bend			0			Type		
49	Bypass seal			Tube-tubesheet joint			Expanded only (2 grooves)(App.A'i')		
50	Expansion joint			Type None					
51	RhoV2-Inlet nozzle 411			Bundle entrance 66			Bundle exit 146		
52	Gaskets - Shell side			Tube side			Flat Metal Jacket Fibe		
53	Floating head								
54	Code requirements			ASME Code Sec VIII Div 1			TEMA class R - refinery service		
55	Weight/Shell 2931			Filled with water 3629.7			Bundle 1523.2		
56	Remarks								
57									
58									

Figure 6 – TEMA Sheet for E-201

Figure 7 – LNG Equipment Capital Cost

UTILITIES: OPERATING COST							
	Project:	Assignment #6	Chemical Engineers:	Khalll Shehadeh	Esteban Vazquez-Elizondo		
	Location:	Chemical Engineering Department	Date:	12/2/2019	Yusef Sorunke		
Utility Class	ASPEN ID	Service	Tag	Usage	Unit Cost	Annual Cost, \$/yr	
Cooling Water		Cooling of Heat Exchangers		217.19 kW	0.354 \$/GJ	\$21,357.02	
Power		Powering compressors and heat exchangers		144.79 kW	16.8 \$/GJ	\$675,686.71	
LNG Product		Selling Product			480\$/1000kg	\$1,774.109	
					Total Revenue	\$1,077,065	

Figure 8 – Claude Model Operating Cost

UTILITIES: OPERATING COST						
	Project:	Assignment #6	Chemical Engineers: Khaili Shehadeh		Esteban Vazquez-Elizondo	
	Location:	Chemical Engineering Department	Date:	12/2/2019	Yusef Sorunke	
Utility Class	ASPEN ID	Service	Tag	Usage	Unit Cost	Annual Cost, \$/yr
Cooling Water		Cooling of Heat Exchangers		299.56 kW	0.354 \$/GJ	\$29,456.73
Power		Powering compressors and heat exchangers		199.71 kW	16.8 \$/GJ	\$931,980.07
LNG Product		Selling Product			480\$/1000kg	\$1,408,199
Total Revenue						\$446,762

Figure 9 – Linde Model Operating Cost

References

Dr. Michael Caracotsios CHE433 Notes

CHE433 Operating Cost Calculator

Marshall and Swift Cost Index, M&S

Aspen Plus V9 Simulator